

Comparative Glossary

content/generated/markdown/textbook/reading-copies/comparative-glossary.md

Comparative Glossary

This glossary provides a comparative reference between standard modern frameworks and the canonical terminology used in AAA.

Term	GR/QM/ Λ CDM	AAA = Architrino Assembly Architecture
Absolute Frame / Preferred Frame	No preferred inertial frame; all inertial frames are equivalent in SR/GR.	The Euclidean-void rest frame defined by isotropy of c_f and absolute time; physically real but operationally hidden.
Absolute Time	No universal time; GR uses coordinate time and proper time along worldlines.	Global, linear time parameter t indexing each $\mathbb{U}_{\text{now}}$ slice; universal synchronization.
Absolute Timespace	No direct equivalent in GR, where spacetime is dynamical and not a fixed product background.	Formal AAA name for the fixed ontological background $\mathbb{R} \times \mathbb{R}^3$ (absolute time + Euclidean void). Use when the full product structure matters. Prefer Euclidean void and absolute time when the prose is naming the spatial and temporal factors directly.
AdS Interior / AdS-Like Region	Anti-de Sitter space is a specific constant-curvature spacetime in GR/string theory.	Certified self-hit branch with confining/steep potential behavior; super-field-speed history is a warning regime, but the branch is accepted only by same-source root existence plus transversality.
Assembly	No exact single standard term; overlaps partially with particle, bound state, composite, or excitation depending on context.	Canonical AAA ontological term for a localized architrino configuration, from a single architrino up through bound hierarchical structures such as binaries, nested shell swarms, and higher composites. Use assembly when the text means the object itself rather than only its observer-level phenomenology.
Architrino	Not part of standard GR/QM/ Λ CDM; no such fundamental entity.	Point transceiver of polarized potential with definite polarity and specific dynamics. In AAA this is the ontological primitive; electric-charge bookkeeping is a downstream effective mapping, not the primary definition.
Axial Architrino	No direct standard-model term; the closest analog would be a constituent carrying part of a fermion's outward electroweak pattern.	One of the six non-core architrimos bound to a fermion assembly's axial organization. Axial architrimos occupy polar sites, contribute to the assembly's visible electroweak bookkeeping, and remain distinct from the neutral Noether swarm scaffold.

Term	GR/QM/ Λ CDM	AAA = Architrino Assembly Architecture
Axial Frame	No direct analogue as a standard named internal frame in particle physics.	The coarse-grained orientation determined by the six-site axial arrangement. It may align with or rotate relative to the fixed Noether swarm frame, and this relative orientation is the preferred language for weak-mixing and misalignment discussion.
Axial Layer	No direct standard term; roughly analogous to an internal electroweak-structure layer of a composite model.	The six-site organization formed by the six axial architrinos on a fermion assembly. Use as the canonical system-level term for fermion axial structure.
Axial Pattern	No direct standard equivalent; closest would be an internal constituent arrangement governing effective quantum numbers.	The sign/layout arrangement of the six axial architrinos across the three core axes. It is the preferred term for the assembly's constituent layout and electroweak patterning.
Axial Inventory	No direct standard equivalent beyond constituent bookkeeping.	The count/composition of axial architrinos in a given assembly or branch. Use for ledger and catalog language.
Binary	Bound two-body system; modeled by forces/fields (e.g., EM or gravity), with no intrinsic delay-isochron mechanics.	Two-body bound system in general (including macroscopic binaries such as neutron-star binaries), plus the architrino-specific base assembly: an opposite-polarity electrino:positrino pair whose principal circular partner branch is anti-damped and whose stabilization, if realized, requires multi-root, self-hit, or wake-energy closure.
Binary, Maximal Curvature (MCB)	No analogue in standard models.	Electrino:positrino pair stabilized at minimum radius via self-hit in $v > c_f$ regime; defines prototype length/time.
Binary, Inner	No specific inner-binary layer; bound systems are not organized into nested binary tiers.	Self-hit (maximal-curvature) binary in the nested shell swarm; highest curvature and smallest radius.
Binary, Middle	No middle-binary tier in standard models.	Field-speed binary at $v = c_f$ acting as energy-storage fulcrum; its radius and frequency are set by the branch geometry.
Binary, Outer	No outer-binary tier in standard models.	Slowest, largest binary; mediates external coupling and approaches $v = c_f$ near alignment.
Nested Shell Swarm	No nested shell swarm assembly; bound states described by quantum fields and forces.	Three nested binaries (inner, middle, outer) with separated radii and frequencies; building block for particles.
Oblate Spheroid / Ellipsoid	An oblate spheroid is an axisymmetric ellipsoid with two equal equatorial semiaxes and one shorter polar semiaxis; ellipsoid is the broader class and may be triaxial.	The Noether swarm geometry should be named as an oblate spheroidal envelope, or oblate spheroidal exclusion envelope when packing/exclusion is active. Use ellipsoid only for generic geometry, comparison families, or genuinely triaxial cases. The Lorentz content belongs in the separate closure law $\xi = R_{\parallel}/R_{\perp} \rightarrow 1/\gamma$, not in a surface name such as Lorentz spheroid.
Causal Cone (c_f)	Light cones define causal influence at speed c .	Causality is defined by finite wake speed c_f in absolute time; cones are built from $r = c_f(t - t_0)$.

Term	GR/QM/ Λ CDM	AAA = Architrino Assembly Architecture
Causal Interaction Set ($\mathcal{C}_{o'j}(t)$)	No direct standard named object; related to delayed-time root sets in classical field theory.	Set of emission times from source j whose causal wakes intersect receiver o' at time t . Source identity fixes the hit type: $j = o'$ gives self-hit, while $j \neq o'$ gives partner hit. Root count is separate: either source can have one or more active roots on a given branch chart.
Causal Isochron	Delayed-time/light-cone surfaces are mathematical constructs; fields are not defined as isochron surfaces.	Expanding causal surface from a single emission time: $r = c_f(t - t_0)$ with $1/r^2$ surface density; intersections deliver radial hits.
Causal Wake	Causal influence propagates along light cones; described by fields/wavefronts.	The causal-isochron residue left by architrino motion; the architrino-native description of what appears as a field at the effective level. More concretely, it is the union of the causal isochrons emitted by a source.
Field (when used as a default AAA substrate term)	Fundamental or effective dynamical object in classical field theory / QFT.	Avoid as the default ontological term for emitted architrino influence. Prefer wake or causal wake for substrate-level AAA prose. Keep field for QFT comparison, effective continuum description, or observer-level translation when that is the real topic.
CFT Region / CFT Space	Conformal field theories are specific QFTs without intrinsic length scales.	Partner-only, sub-field-speed regime ($v < c_f$) associated with exterior/bulk behavior.
Clock-Rate Redshift	Cosmological redshift is usually modeled by metric expansion with $1 + z = a_{\text{obs}}/a_{\text{emit}}$.	Redshift is interpreted as medium-evolution plus path-integrated clock-rate comparison between emission and observation environments in a fixed Euclidean void.
Charge	Usually a point-particle or field-source property such as electric, color, or gauge charge.	Use with care. At the architrino level the preferred ontology is not “point charge” but an architrino as a transceiver of potential with definite polarity and specific dynamics. Reserve charge primarily for historical comparison, continuity with standard physics, or higher-level effective/bookkeeping roles where charge language is the correct abstraction.
Deterministic Multistability	Standard models usually present either deterministic evolution equations or probabilistic outcomes; this combined term is not standard.	Laws are deterministic, but near threshold/self-hit regimes multiple stable attractors can coexist; realized outcomes are microstate-sensitive without ontic randomness. Non-Markovian.
Effective Metric	Metric $g_{\mu\nu}$ is fundamental in GR, not an emergent medium property.	Emergent refractive geometry from Noether sea density/stress that reproduces GR-like behavior.
Noether sea Delay Factor (χ_{sea})	Refractive index n in ordinary optical-medium analogies.	Dimensionless delay factor $\chi_{\text{sea}}(\mathbf{x}, t) = c_f/c_{\text{eff}}(\mathbf{x}, t)$ measuring effective causal slowing through the Noether sea. Use this instead of optical n so normalized Noether swarm density keeps the symbol n .
Effective Scale Factor	In standard cosmology, $a(t)$ is the FLRW scale factor describing metric expansion of space.	$a(t)$ is an effective summary of Noether sea structural evolution (for example via representative assembly-separation scales) inside a fixed Euclidean container.
Electrino	No fundamental $-e/6$ charge; electron is elementary with charge	Negative-polarity architrino. When electric bookkeeping is needed for comparison or continuity, it carries effective label

Term	GR/QM/ Λ CDM	$\mathbb{A}\mathbb{A}\mathbb{A}$ = Architrino Assembly Architecture
	$-e_*$	$q = -\epsilon$ where $\epsilon = e /6$.
Emission Time	Usually implicit as a delayed source time in propagators/Green-function methods.	Explicit source-emission timestamp t_0 for a wake contribution later received at $(\mathbf{x}, t)$ under the causal condition.
Euclidean Void	No fixed Euclidean background; GR uses dynamical spacetime geometry, QM/QFT assumes a background only in specific approximations.	Continuous, unmoving $\mathbb{R}^3$ container.
Background ($\mathbb{A}\mathbb{A}\mathbb{A}$ ontology)	Often ambiguous between spacetime, vacuum, and medium.	Prefer background for the fixed container only: absolute time + Euclidean void. Do not let it silently absorb the Noether sea contents.
Event-Horizon Alignment Condition	Event horizon is a null surface in GR.	Planck-scale alignment: middle and outer binaries reach $v = c_f$ and become coplanar/co-linear with the inner binary; precession ceases.
Field Speed (c_f , $\mathbb{A}\mathbb{A}\mathbb{A}$ reference speed)	Speed of light c in vacuum; invariant signal speed in SR/GR.	Propagation speed of architrino potential wakes relative to the void; reference speed for self-hit threshold. Speed of light is not constant but approaches c_f in a low-energy Noether sea, i.e., deep space with only minimal gravitational gradient.
Field-Speed Threshold	No special threshold at $v = c$ beyond kinematic invariance.	Symmetry-breaking boundary at $v = c_f$ separating partner-only and self-hit regimes.
Gauge Symmetry	Local redundancy or invariance of a field description; in the Standard Model the internal gauge symmetries organize the observed interaction sectors.	Effective invariance of the observer-level record under local changes of bookkeeping basis. It is a closure target for assembly, causal-wake, axial-layer, and Noether sea behavior, not a primitive substrate ontology.
Gauge Group / Standard Model Gauge Record	The Standard Model internal gauge structure $U(1)_Y \times SU(2)_L \times SU(3)_c$, together with representations, charges, couplings, anomaly cancellation, and mixing data.	The low-energy effective gauge record that $\mathbb{A}\mathbb{A}\mathbb{A}$ must recover from assembly geometry and medium response. The accepted record must reproduce the gauge bookkeeping while keeping extra low-energy partners, transport modes, and instability channels absent.
Gauge Field / Gauge Connection	Connection-like field variables such as A_μ , W_μ^a , and G_μ^a that compensate local gauge-basis changes and enter covariant derivatives.	Effective connection variables reconstructed from the causal-wake/action ledger and Noether sea or assembly state. Use them for observer-level phase, weak-basis, or color-basis transport; do not promote them to independent substrate fields.
Gauge Curvature / Field Strength	Curvature of a gauge connection, usually written as field-strength tensors; it carries measurable force, flux, self-interaction, and transport content.	Observer-level curvature or holonomy summary recovered from the same effective gauge record used for force and phase transport. It measures failure of the effective connection to cancel around loops or across regions, read as force, phase shift, flux, weak-channel response, or color transport.
Gauge Covariance	Requirement that equations keep the same physical content under local	Requirement that effective equations preserve the same physical record under local changes of phase, weak, or color basis. It fails if medium-response, preferred-frame, or branch-

Term	GR/QM/ Λ CDM	$\mathbb{A}\mathbb{A}\mathbb{A}$ = Architrino Assembly Architecture
	gauge transformations, even though the local field components change.	record corrections force leading-order gauge-breaking operators in the validated regime.
Gauge Coupling Constants (g_1, g_2, g_3)	Parameters controlling the strengths of the hypercharge, weak, and color interactions; in QFT they run with probe energy under renormalization.	Scale-dependent exposure and response coefficients to be derived as shared outputs of branch geometry, axial-layer exposure, color axis-exceptionality, and Noether sea response. They are not per-observable fit constants.
Gauge Holonomy / Gauge Topology	Gauge-invariant information carried by parallel transport around loops, flux sectors, instanton-like sectors, or chart-gluing data.	Observer-level loop or sector record extracted from the retained wake/action ledger. Aharonov-Bohm phase, flux compatibility, and topological sector labels must be read from one shared effective gauge record rather than inserted as independent charges.
Gauge Symmetry Breaking / Higgs Mechanism	Mechanism by which a higher-symmetry gauge description yields lower-energy observed sectors; in the electroweak theory the Higgs field and nonzero vacuum expectation value set masses and mixing.	Effective phase-change and medium-response closure target. The order parameter, electroweak normalization, and mass/mixing outputs must be derived from Noether sea state and assembly variables rather than treated as evidence that the Euclidean void has its own field density.
Gauge Unification / Grand Unified Theory	Larger symmetry containers that embed the Standard Model gauge group and often motivate proton-instability, partner, or extra-boson predictions.	Comparison framework only until one shared branch record recovers the Standard Model gauge record, reproduces running couplings, and explains the absence of mirror matter, superpartners, proton-instability channels, extra gauge bosons, hidden transport modes, and other non-baseline outputs.
Running Couplings	Renormalization-group scale dependence of interaction strengths, often used as evidence pressure for high-energy unification.	Scale-dependent effective gauge record $\mathbf{g}_{\mathbb{A}\mathbb{A}\mathbb{A}}(\mu; \theta)$ recovered through the gauge-running residual. Near-convergence at high scale is a comparison diagnostic, not proof of fundamental gauge unification.
$SU(3)_c$ Color Sector	Non-Abelian color gauge sector of QCD; quarks transform as color triplets, gluons as the adjoint octet, and confinement governs observable color singlets.	Effective color transport from Noether swarm axis exceptionality and axial-layer bookkeeping. The color algebra closure target is to recover an $\mathfrak{su}(3)$ generator set and confinement-facing transport without treating gluon fields as primitive substrate objects.
$SU(2)_L$ Weak Sector	Left-handed weak-isospin gauge sector; weak doublets transform under $SU(2)_L$, while right-handed charged fermions are singlets in the Standard Model.	Effective weak-basis covariance of the exposed weak-coupling-triad channel. The native burden is to derive left-channel exposure, right-channel suppression, weak mixing, CKM/PMNS overlap, and weak-reaction provenance from one weak-exposure domain.
$U(1)_Y$ Hypercharge Sector	Abelian weak-hypercharge gauge sector; after electroweak mixing it combines with $SU(2)_L$ to yield electromagnetism and the Z channel, with $Q = T_3 + Y/2$.	Effective hypercharge and electric-bookkeeping record carried by shielded axial inventory, weak-coupling-triad state, and core-offset bookkeeping. It is the observer-level charge relation to be recovered from assembly structure, not a primitive architrino charge ontology.

Term	GR/QM/ Λ CDM	AAA = Architrino Assembly Architecture
Hybrid Regime	Not a standard single-term category in Λ CDM; mixed models exist across modified-gravity and dark-matter literature.	Both together; neutral assemblies provide baseline mass, medium response modulates behavior by scale/environment.
Horizon Interface	Not a standard term in GR/QM/ Λ CDM as used here.	Interface layer where middle and outer binaries lock at $v = c_f$; surface degrees of freedom at Planck alignment.
Medium Response	In standard cosmology this is often represented through effective modified-gravity response or environment-dependent phenomenology.	Gravity-like effects from Noether sea elasticity/response, not extra matter alone.
Measurement Interaction	In standard usage, “measurement” is often formalized by projection, POVMs, or decoherence models.	Physical interaction between assemblies and environment/detector that drives irreversible record formation and effective outcome selection without invoking ontic wavefunction collapse.
Neutral Assemblies	Rough analog to non-luminous gravitating matter components in dark-sector modeling.	Dark-sector-like ‘gravitating’ structures made of neutral Noether swarm assemblies. Gravity is the result of local and bulk deformation of Noether sea assemblies.
Noether Swarm	No such assembly; vacuum is not composed of swarms.	General class of Noether-neutral swarm assemblies; reader-facing subtypes are neutral swarm, shell swarm, and nested shell swarm. In Noether sea prose, a Noether swarm usually means the neutral medium assembly under discussion.
Noether sea	No physical sea; vacuum is the ground state of fields, not a substance.	Lattice/network/clusters of Noether swarms occupying the Euclidean void; the canonical ontological term for the ambient substrate contents.
Noether sea Density (ρ_{NS}, n)	Vacuum energy density or cosmological-constant density ρ_Λ ; typically treated as constant in Λ CDM.	Local Noether swarm density $\rho_{NS}(\mathbf{x}, t)$ and normalized density $n(\mathbf{x}, t) = \rho_{NS}/\rho_{NS,0}$. Density helps determine clock rates and the Noether sea delay factor χ_{sea} , but it is not itself the delay factor.
Non-Markovian Memory	Memory effects appear in many-body/open-system models but are not a foundational ontology term in GR/QM/ Λ CDM.	Evolution depends on path history (past wake emissions/intersections), not only the instantaneous state; key in self-hit dynamics and delayed interactions.
Multi-Root Regime	Multiple delayed source times may contribute to a field value, but this is not usually treated as a primitive ontology.	Regime in which a receiver has more than one active causal root at the same reception time. Multi-root is a root-count term, not a source-identity term: it may include multiple partner-hit roots, multiple self-hit roots, or both.
Partner Hit	No analogue as a discrete wake-intersection primitive; closest comparisons are delayed two-body field contributions.	Hit whose source is another architrino ($j \neq i$). In an electrino:positrino binary, partner hits are attractive and supply the inward channel. Multiple partner hits are possible at higher speeds when older partner wake surfaces also satisfy the causal-root condition.

Term	GR/QM/ Λ CDM	$\mathbb{A}\mathbb{A}\mathbb{A}$ = Architrino Assembly Architecture
Partner-Only Regime / Partner-Only Hit Regime	No analogue in standard models.	Regime in which a receiver intercepts partner wakes but not its own wake. In the simple circular binary this is the sub-field-speed side of the field-speed threshold. Prefer partner-only regime in prose and reserve partner hit for an individual received contribution.
Particle	Localized excitation, detector event, or asymptotic object in standard quantum and particle-physics language.	Keep mainly as an observer-level, experimental, or Standard Model comparative term. In native $\mathbb{A}\mathbb{A}\mathbb{A}$ ontology, what is often called a particle is usually an assembly together with its emitted wake structure.
Particle-Wave Duality	Standard quantum tension in which the same object exhibits particle-like localization and wave-like interference.	Not treated as a fundamental ontological paradox. $\mathbb{A}\mathbb{A}\mathbb{A}$ responds with assembly and wake: the assembly is the localized source-organized object, while the wake is the distributed causal structure that propagates, overlaps, and interferes.
Path History	In quantum theories, typically framed as Feynman path contributions of Standard Model particles/fields in amplitudes and propagators.	Standard Model particles are assemblies of architrinos; the fundamental path-history object is the set of architrino trajectories and their causal wake-emission history.
Planck Scale ($\mathbb{A}\mathbb{A}\mathbb{A}$ usage)	Scale set by $G, \hbar, c$ (Planck length/time/energy) where quantum gravity is expected.	Event-horizon alignment condition where the middle and outer binaries reach $v = c_f$ and become coplanar/co-linear with the inner binary.
Polar Dyad	No standard-model equivalent as a named structural unit.	The two polar sites associated with one Noether swarm axis. A polar dyad is the preferred local geometric term for the site pair on a single axis.
Polar Site	No direct standard-model equivalent.	One of the six local attachment/dwelling positions of a nested shell swarm assembly occupied by the axial architrinos. Use polar for site geometry and axial for the six-site system-level organization.
Positrino	No fundamental $+e/6$ charge; positron is the electron antiparticle with charge $+e$.	Positive-polarity architrino. When electric bookkeeping is needed for comparison or continuity, it carries effective label $q = +\epsilon$ where $\epsilon = e /6$.
Potential Fluctuation	Typically treated as field/vacuum fluctuations in QFT language.	Context-dependent $\mathbb{A}\mathbb{A}\mathbb{A}$ term for transient, local potential-pattern variation from superposition in the architrino/Noether sea system; use when “virtual particle” language would be misleading.
Proper Time	Invariant time along a worldline; for signature $(-, +, +, +)$, $d\tau^2 = -g_{\mu\nu}dx^\mu dx^\nu$; clock time in GR.	Time measured by physical clocks (Noether swarm assemblies) interacting with the Noether sea; emergent from dynamics.
Provenance / Provenance Ledger	No direct analogue as a mandatory standard output term in GR/QM/ Λ CDM.	Bookkeeping records for each received contribution (e.g., emitter identity, emission time, receiver/time index, contribution strength); required for simulation validation and causal auditing.

Term	GR/QM/ Λ CDM	AAA = Architrino Assembly Architecture
Reaction	Usually a broad process label for scattering, transformation, decay, fusion, or rearrangement channels.	Preferred broad AAA event term when the prose names a channel without yet specifying the exact mechanism. Use as the native replacement for generic Standard Model decay wording in textbook-facing prose. A reaction may be multi-assembly, medium-mediated, or apparently spontaneous when a single assembly (immersed in the Noether sea) reaches an instability threshold and undergoes causal reconfiguration.
Spontaneous pair production	Pair creation in a strong field or unstable vacuum without an incoming particle directly triggering the event.	Spontaneous assembly-pair production from a local Noether Pair or equivalent neutral source architecture. In AAA, spontaneous does not mean uncaused; it means the observer-level description does not resolve the local micro-causal trigger.
Reaction, Beta	Standard weak nuclear process historically labeled beta decay.	Preferred AAA native label for that channel. At first mention one may write beta reaction (SM label: beta decay), then continue with beta reaction.
Refractive Gravity	No standard term as a foundational ontology in GR/QM/ Λ CDM; gravity is modeled as spacetime curvature in GR.	AAA interpretation in which gravity-like effects arise from propagation and clock/ruler response in a non-uniform Noether sea (effective refractive medium), yielding GR-like phenomenology without fundamental void curvature.
Root Ledger / Causal-Root Ledger	No direct standard object; roughly analogous to branch bookkeeping in multi-delay equations.	Integer bookkeeping of active causal roots on a fixed branch chart. In circular binary notation, N_s counts active self-hit branches and M_p counts active partner-hit branches. The ledger records admissible delayed channels; it is not itself a force law or a stability proof.
Self-Hit	No self-intercepting field-wake dynamics in standard GR/QM.	Hit whose source and receiver are the same architrino ($j = i$): an architrino intercepts its own causal wake. In the simple circular binary the principal self-hit branch turns on at $v > c_f$, creating non-Markovian feedback and the candidate outward channel that can oppose collapse. In general, self-hit is accepted by same-source root existence plus transversality, not by speed alone.
Self-Hit Onset Boundary	No analogue in standard models.	Branch boundary at which the same-source causal-root set becomes nonempty with a valid transversality/Jacobian floor. The scalar condition $v > c_f$ is a regime warning and a necessary ingredient for simple nontrivial roots over the intervening interval, not a sufficient criterion by itself.
SMBH-Core Recycling	SMBHs in standard cosmology are compact objects in evolving spacetime, with accretion/feedback but no required cosmological recycling ontology.	SMBH interiors and horizon channels act as long-lived source/release regions that drive recurrent medium-loading, relaxation, and large-scale cosmological history in the Architrino framework.

Term	GR/QM/ Λ CDM	$\mathbb{A}\mathbb{A}\mathbb{A}$ = Architrino Assembly Architecture
Spacetime Medium	No physical medium; spacetime is the metric field and vacuum is not a substance.	Pervasive assembly network giving rise to effective metric/inertia.
Spacetime	Fundamental geometric arena in GR; in relativistic/QFT practice it is the default background or dynamical metric structure.	Use for emergent effective geometry, relativistic observer-level description, or coarse-grained metric language. Do not use as the default $\mathbb{A}\mathbb{A}\mathbb{A}$ name for the ontological substrate, which is instead articulated as absolute time + Euclidean void + Noether sea.
Spiral	Orbital spirals arise from radiation reaction, dissipation, or non-conservative forces; not a fundamental mechanism.	In $\mathbb{A}\mathbb{A}\mathbb{A}$, a logarithmic-like inward spiral is a non-circular branch ansatz or capture target. It is not derived from the principal partner circular branch, whose positive tangential work is anti-damping.
Super-Field-Speed Regime	Superluminal particle motion is generally forbidden in special-relativistic kinematics.	Regime in which an architrino or internal assembly component has speed $v > c_f$ relative to the Euclidean void. This does not mean wakes propagate faster than c_f ; it means the moving source can outrun its own earlier wake surfaces and become a candidate for same-source roots. Prefer super-field-speed over supra-field-speed.
Transient Assembly	No single direct counterpart; roughly overlaps with short-lived intermediate states in QFT language.	Short-lived, non-asymptotic architrino configuration that associates and later dissociates during interactions; preferred $\mathbb{A}\mathbb{A}\mathbb{A}$ replacement for ambiguous “virtual particle” phrasing.
Nested Shell Swarm Alignment Plateau	No analogue in standard models.	Locked alignment state reached as total velocity approaches c_f ; planes align and frequencies lock at integer ratios.
$\mathbb{U}_{\text{now}} \equiv S(t)$	No direct equivalent as a fundamental ontic slice in GR/QM/ Λ CDM; practical descriptions are observer/model-dependent.	Complete ontic universe state on simultaneity slice Σ_t at global time t : $S(t) = \{(\mathbf{x}_i(t), \mathbf{v}_i(t), q_i, \dots)\}$, together with path-history/provenance bookkeeping (emitter identity, emission time) and self-hit history needed for deterministic evolution. Here $\mathbb{U}$ denotes Universe. This is simulation ground truth. Not an observer.
Wave	Distributed oscillatory or amplitude-carrying structure in classical or quantum language; often part of wavefunction or field descriptions.	Use with care. In $\mathbb{A}\mathbb{A}\mathbb{A}$ substrate ontology, the preferred term for emitted causal structure is wake, not wave. Keep wave for standard QM comparison, observer-level phenomenology, or explicitly effective descriptions where that is the intended level.
Wave Function Transition	In standard QM language, state changes are described via unitary evolution plus measurement update/collapse formalisms, e.g., “collapse of the wave function”.	$\mathbb{A}\mathbb{A}\mathbb{A}$ term for effective state-description transition during measurement interaction and decoherence-driven record formation; the underlying architrino dynamics remain continuous and ontic (no fundamental collapse postulate).

Term to Avoid / Phase Out	Transitional Wording	Preferred AAA Wording	Usage Note
vacuum (as physical emptiness)	vacuum state / medium context	Noether sea; spacetime medium	Use “vacuum” only when explicitly discussing mainstream historical semantics.
ether / aether (alone)	medium (clarified)	Noether sea; spacetime medium	Avoid ambiguous historical “luminiferous ether” connotations.
background (when meaning the fixed container)	background spacetime	Euclidean void; absolute background	Keep background tied to the fixed container in ontology-first prose.
background (when meaning what fills the container)	medium / vacuum background	Noether sea; ambient substrate contents; spacetime medium	Do not blur the contents with the fixed background itself.
curved space (as fundamental ontology)	effective geometry	effective metric; refractive geometry from medium density/stress	In AAA, fundamental ontology is Euclidean void + assembly medium; GR-like curvature is emergent.
delayed source-time shorthand	delayed causal contribution	path history; causal wake surface; causal isochron	Keeps language tied to explicit wake-emission bookkeeping.
field (for emitted architrino influence)	effective field	wake; causal wake	Prefer wake for the emitted substrate structure. Keep field only when the discussion is explicitly effective, continuum, or QFT-comparative.
wake field	mixed hybrid wording	causal wake; effective field	Split the phrase according to level instead of keeping the hybrid.
decay (as generic process wording in AAA prose)	historical Standard Model label; quoted canonical label on first mention only	reaction; dissociation; association; reconfiguration	Do not use decay as the default explanatory verb in AAA prose. Keep it only for fixed Standard Model labels, for example beta decay, then translate immediately into native wording such as beta reaction.
field line (in AAA substrate prose)	line in a continuum field picture	line of action; radial line of action	The canonical AAA interaction law is radial and wake-based rather than line-of-force based.
field shell / field front	wavefront in a field picture	wake surface; wake front; causal isochron	Makes the emitted geometry explicit.
shell (for emissions/intersections/self-hit)	wake front / wake surface	causal wake; causal isochron	Reserve “shell” for unrelated chemistry contexts (e.g., electron shells).

Term to Avoid / Phase Out	Transitional Wording	Preferred AAA Wording	Usage Note
virtual particles (as ontic entities)	effective intermediate in field description	transient assembly configuration	Emphasize assembly-level intermediates rather than ontic virtual quanta.
wavefunction collapse (as fundamental discontinuity)	effective state update under measurement	measurement interaction; wave function transition	AAA framing keeps underlying dynamics continuous/ontic.
charge (when used as the default name for an architrino)	charge-like polarity bookkeeping	architrino; polarity; axial architrino; transceiver of potential	Avoid collapsing architrino ontology into point-charge language. Keep charge for historical continuity, standard-model comparison, or higher-level electric/color/gauge bookkeeping when that is the real topic.
global time (when referring to universal clocking)	universal time parameter	absolute time; universal synchronization	Keep “global” for non-time global constraints (e.g., neutrality) when mathematically precise.
spacetime (when used as a standalone fundamental ontology)	effective/emergent spacetime description	spacetime medium; Noether sea; Euclidean void	AAA uses spacetime at the effective level. Use spacetime medium only as a bridge term, Noether sea for substrate contents, and Euclidean void for the fixed container.
timespace (used loosely)	compressed synonym for spacetime/background	absolute timespace; Euclidean void; absolute time	Keep absolute timespace as the formal product-manifold term only when needed. Avoid loose use of timespace as an all-purpose ontology word.
vacuum fluctuation (if interpreted as empty-space randomness)	effective fluctuation in medium fields	potential fluctuation in architrino/Noether sea system	Use when describing local transient potential-pattern variation.
particle-wave duality (as native ontology)	standard quantum paradox language	assembly and wake	In AAA the localized object and the distributed propagation structure are complementary components of one causal picture, not mutually exclusive identities.

Cosmology Lineage Terms

Term	Meaning in current docs	Scope note
QSSC-like	Eternal/recycling cosmology motif with repeated source history rather than one unique global origin.	Conceptual lineage marker only; not a full ontology import.

Term	Meaning in current docs	Scope note
Cyclical-like	Repeated effective epochs in observer chronology mapping.	Used to orient chronology language; mechanism remains SMBH-local in AAA.
Timescape-like	Environment-conditioned clock calibration affecting inferred background history.	Descriptive bridge only; implemented through explicit Noether sea variables.
Distributed CMB sourcing	CMB source interpretation as long-time accumulated source + transport + thermalization channels.	Must still satisfy blackbody and TT/TE/EE closure constraints.
Tired-light (excluded in AAA core)	Generic scattering-loss redshift hypothesis.	Excluded when it violates image sharpness or $(1+z)$ time-dilation consistency.

Key Regime Terms

Three-column map of AAA terminology across the nested shell swarm regimes.

Aspect	Self-Hit Regime	Symmetry-Breaking Point	Partner-Only Regime
Speed regime	$v > c_f$	$v = c_f$	$v < c_f$
Regime names	Self-hit regime; super-field-speed	Symmetry-breaking point; symmetry-breaking threshold; self-hit onset boundary; field-speed threshold; $v = c_f$ hinge	Partner-only regime; non-self-hit; sub-field-speed
Energy level	High energy binary	Medium energy binary	Low energy binary
AdS/CFT mapping	AdS interior; AdS-like;	Holographic horizon/interface; Horizon interface; event horizon; Planck-scale alignment condition	CFT (exterior QFT); CFT region; CFT/Noether sea
Black hole region	inside a black hole	black hole event horizon	outside a black hole
Geometry/dynamics	“spacetime” inflation/deflation	flat; planar/2D	“spacetime” expansion/contraction
Nested shell swarm binary	Inner Binary (self-hit)	Middle Binary (interface)	Outer Binary (non-self-hit)
Frequency label (tentative hypothesis)	$f = 4x$	$f = 2x$	$f = 1x$